



Biochemistry of Musculoskeletal system

Objective of the session

❖ At the end of this session students will be able:-

- Describe Calcium and bone metabolism.
- Elucidate Collagen biosynthesis, metabolism and interrelated diseases
- Explain Muscle metabolism

Bone

❖ Bone is a mineralized connective tissue composed of various cells and materials.

i. Cellular:

- **Osteoblasts**- Bone forming cells.
- **Osteoclasts**- These cells helps in bone resorption.
- **Osteocytes** – maintain bone, 90%-95% of cells in mature skeleton.

ii. Matrix:

- **Inorganic matrix**= Ca^{++} , Mg^{++} , phosphate, carbonate, chloride, fluoride and citrate.
- **Organic matrix**= mainly type I collagen fibers

Bone cont'd

- ❖ Inorganic content give rigidity to the bone
 - Calcium hydroxyapatite
 - Lack of inorganic content- **soft bone**
- ❖ Organic content give the elasticity to the bone.
 - Commonest is Type 1 collagen,
 - Lack of organic content- **brittle bone**

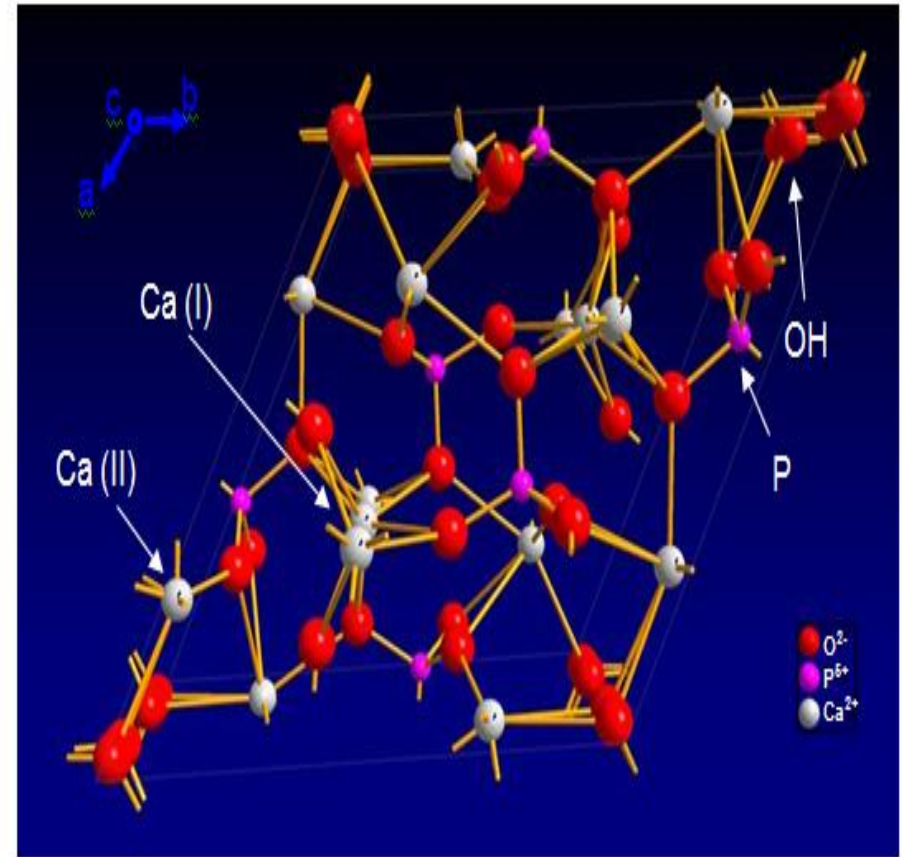
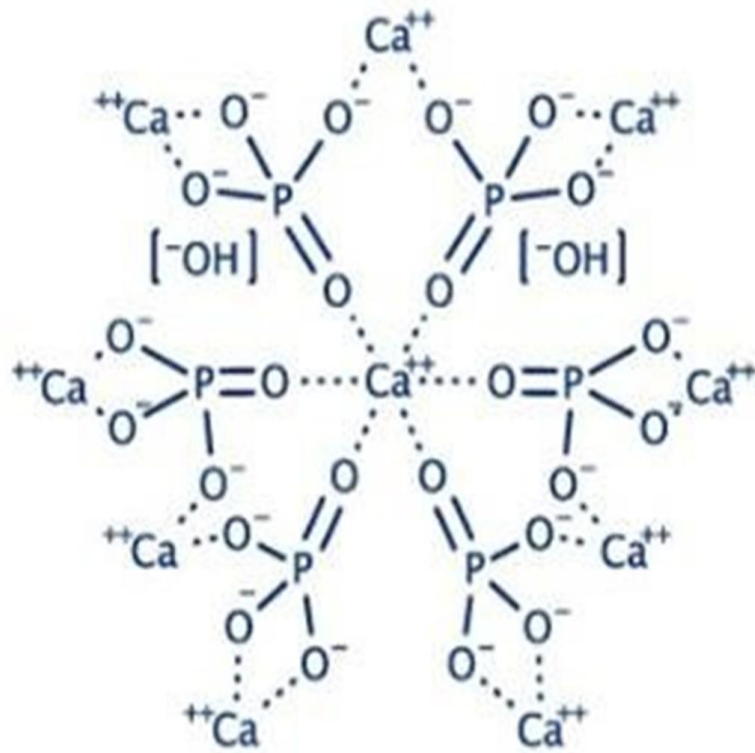
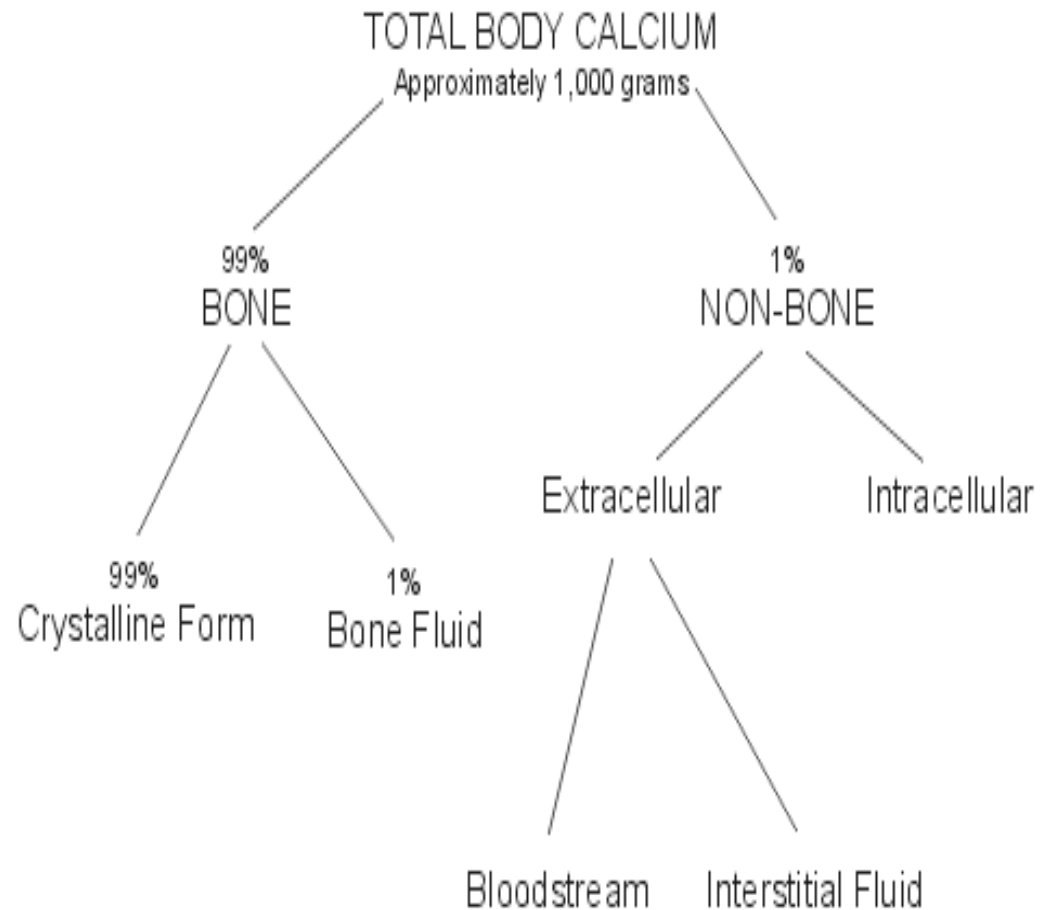


Fig 1. Structure of calcium in the bone:-calcium phosphate (Hydroxyapatite ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$))

Calcium

- ❖ The most abundant mineral in the body
- ❖ ~1.5% of body weight is Calcium,
- ❖ about 1000 -1300 gm
- ❖ ~99% is in the skeleton.

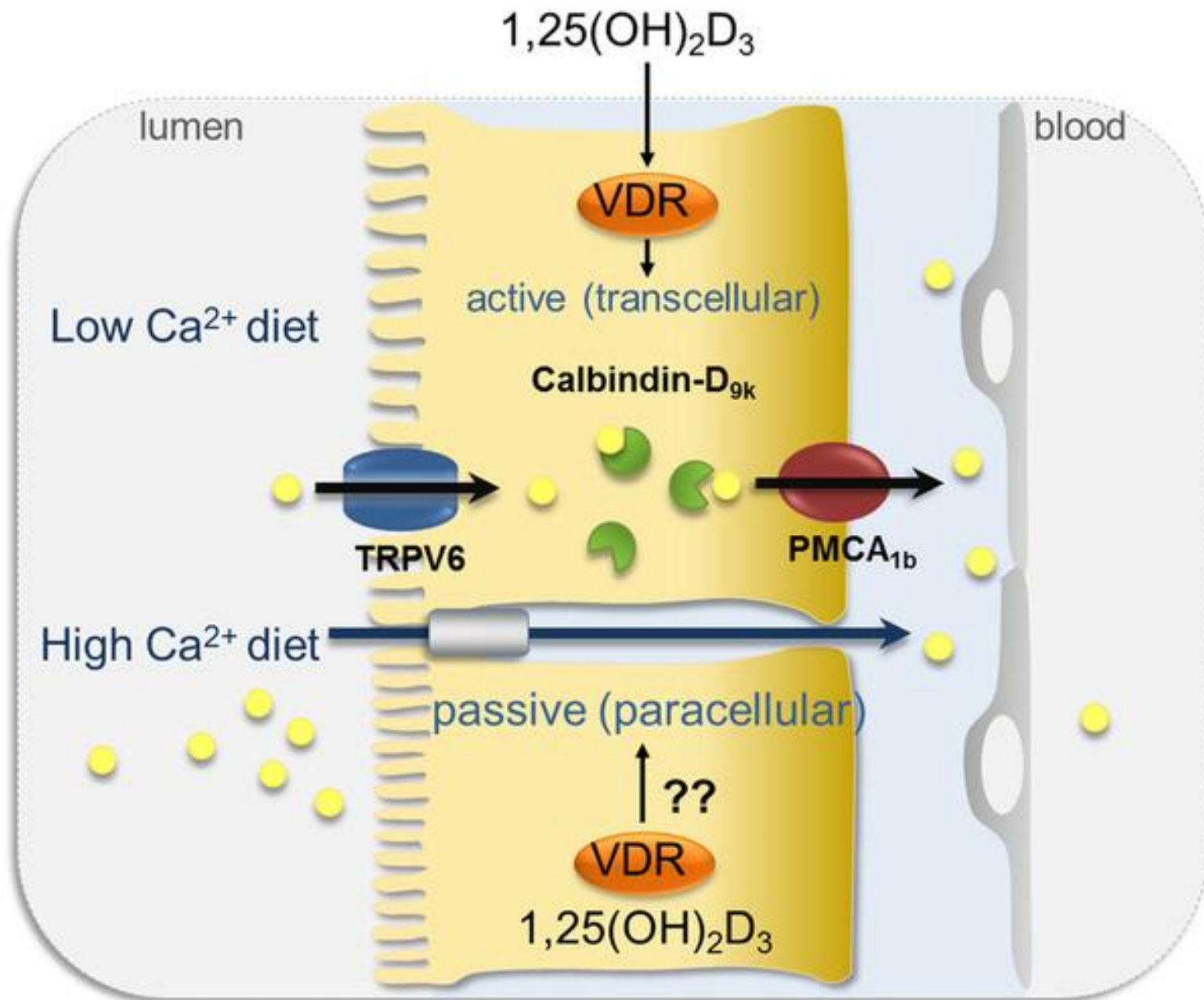


Summary Function of Ca

- i. Bone and teeth mineralization
 - ii. Coagulopathy,
 - iii. Nerve function,
 - iv. Contraction of muscle.
- ❖ Calcium in bone occurs in **two forms** :
- ✓ **Readily exchangeable** and
 - ✓ **Slowly exchangeable** which is more stable form and concerned with bone remodeling by constant resorption & deposition

Absorption of Calcium

- ❖ Ca^{++} from foods is absorbed by both **active transport** and by **passive diffusion** across the intestinal mucosa.
- ❖ Active transport is responsible for most absorption when calcium **intakes are lower**, and
- ❖ Passive diffusion accounts for an increasing proportion of calcium absorption as **intakes rise**.
- ❖ **Vitamin D** is required for
 - Ca^{++} to be absorbed in the gut &
 - to maintain adequate calcium levels in blood



Absorption of Calciumcont'd

- ❖ An inverse relationship between **calcium intake & absorption**.
- ❖ Absorption of calcium from food is
 - about **45%** at intakes of 200 mg/day
 - only **15%** when intakes are higher than 2,000 mg/day.
- ❖ Age can also affect absorption of dietary calcium.
 - as high as **60%** in infants and young children,
 - but it decreases to about **25%** in adulthood
- ❖ **Inhibited:**
 - phytic acid
 - phosphate

Calcium homeostasis

- ❖ refers to the body's **complex process** of maintaining a stable level of Ca^{++} in blood, essential for various bodily functions.
- ❖ Ca^{++} homeostasis is kept by actions of hormones that regulate calcium transport in
 - **gut,**
 - **kidneys &**
 - **bone**
- ❖ The 3 primary hormones are
 - **parathyroid hormone**
 - **Vitamin D3 &**
 - **calcitonin**

Calcium homeostasis.....

- ❖ The primary goal is to keep the calcium concentration in the blood (serum calcium) within a narrow range, typically 8.5 to 10.5 mg/dL.
- ❖ It is maintained through feedback mechanisms, where changes in serum Ca^{++} levels trigger hormone production
 - **decrease** in serum calcium stimulates PTH release,
 - **increase** in serum calcium promotes calcitonin release.
- ❖ Disruptions and Consequences:
 - **Hypocalcemia**: muscle spasms, tetany, and other neurological issues.
 - **Hypercalcemia**: lethargy, confusion, and kidney problems.

Calcium homeostasis.....

1. Parathyroid Hormone (PTH):

- Released by the parathyroid glands,
- **increases blood calcium levels** by
 - stimulating Ca^{++} reabsorption in the kidneys and bone
 - promoting Vit D production $\rightarrow$ $\uparrow$ Ca^{++} absorption GIT

2. Vitamin D:

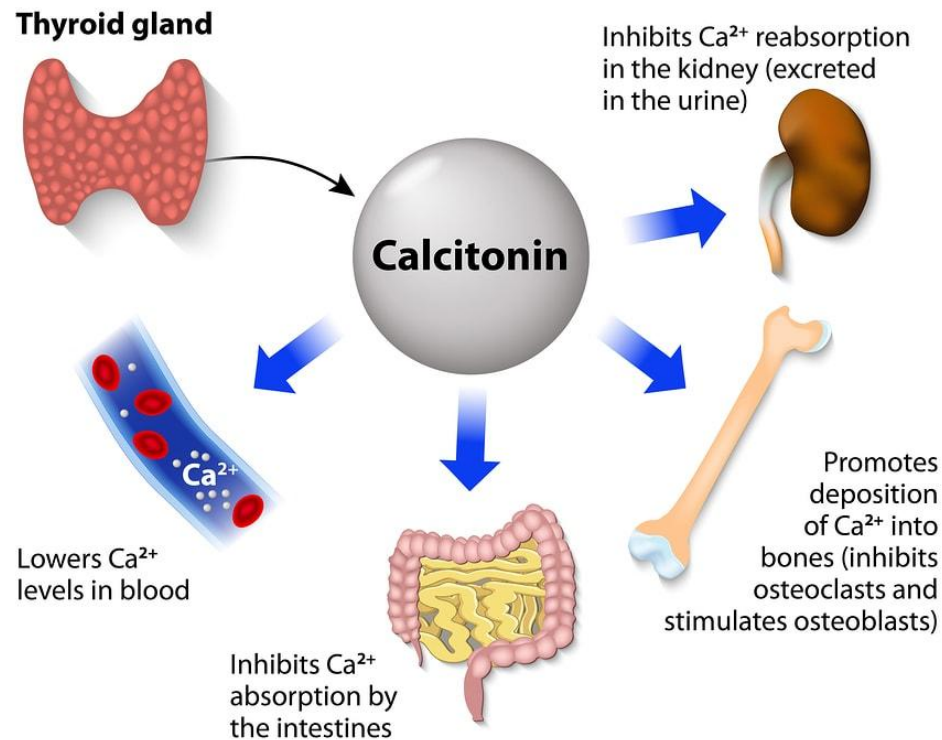
- Activated by the kidneys $\rightarrow$ calcium absorption from GIT.

3. Calcitonin:

- Produced by the thyroid gland,
- **lowers blood Ca^{++} levels** by inhibiting **bone resorption**.

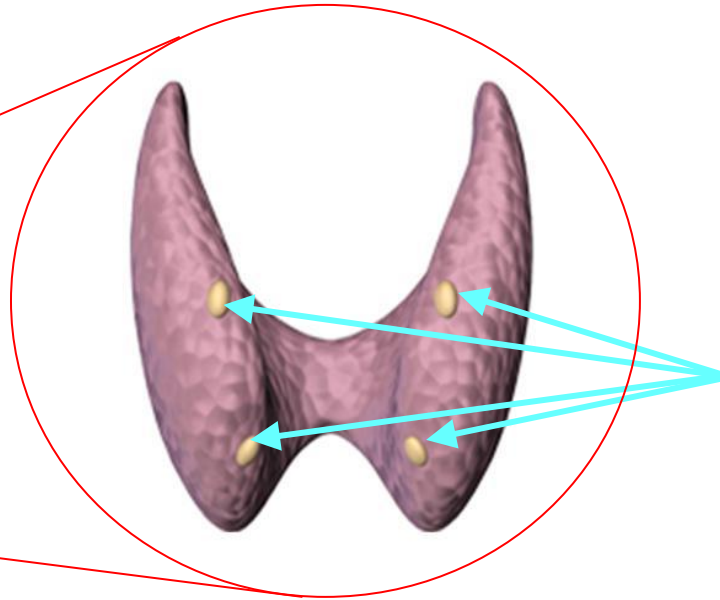
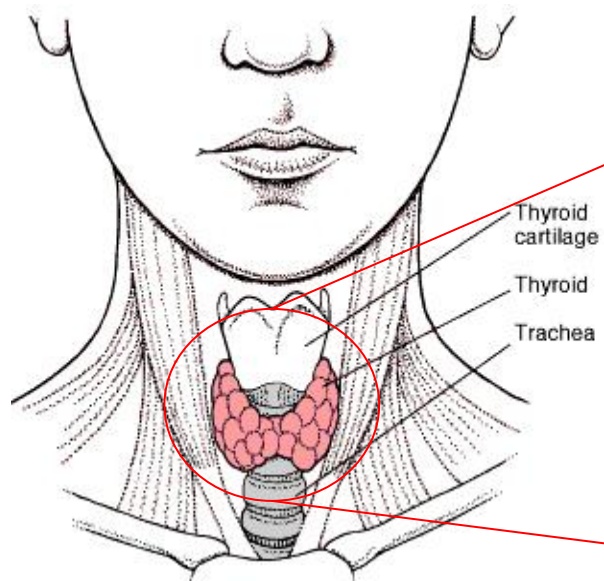
Calcitonin

- ❖ Thyroid C cells
- ❖ Ca^{2+} lowering hormone
- ❖ Used in malignant hypercalcemia, osteoporosis, Paget's disease



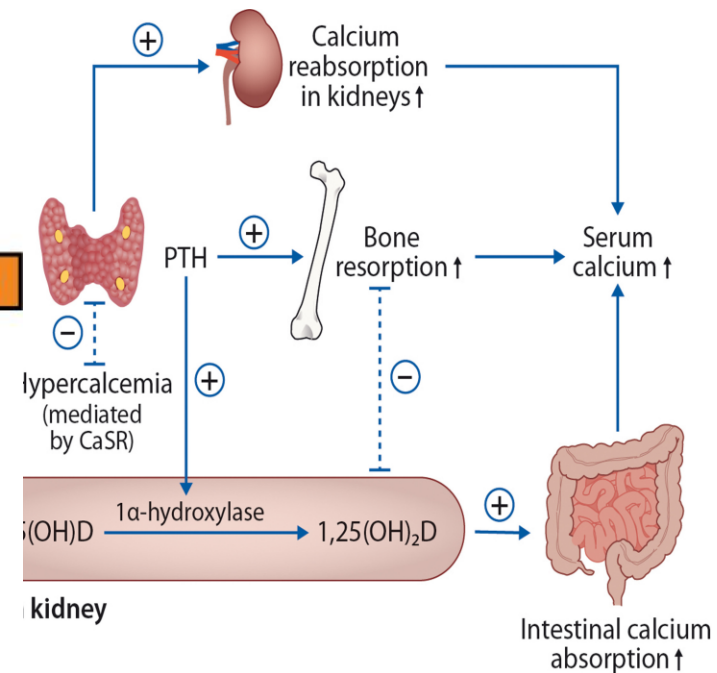
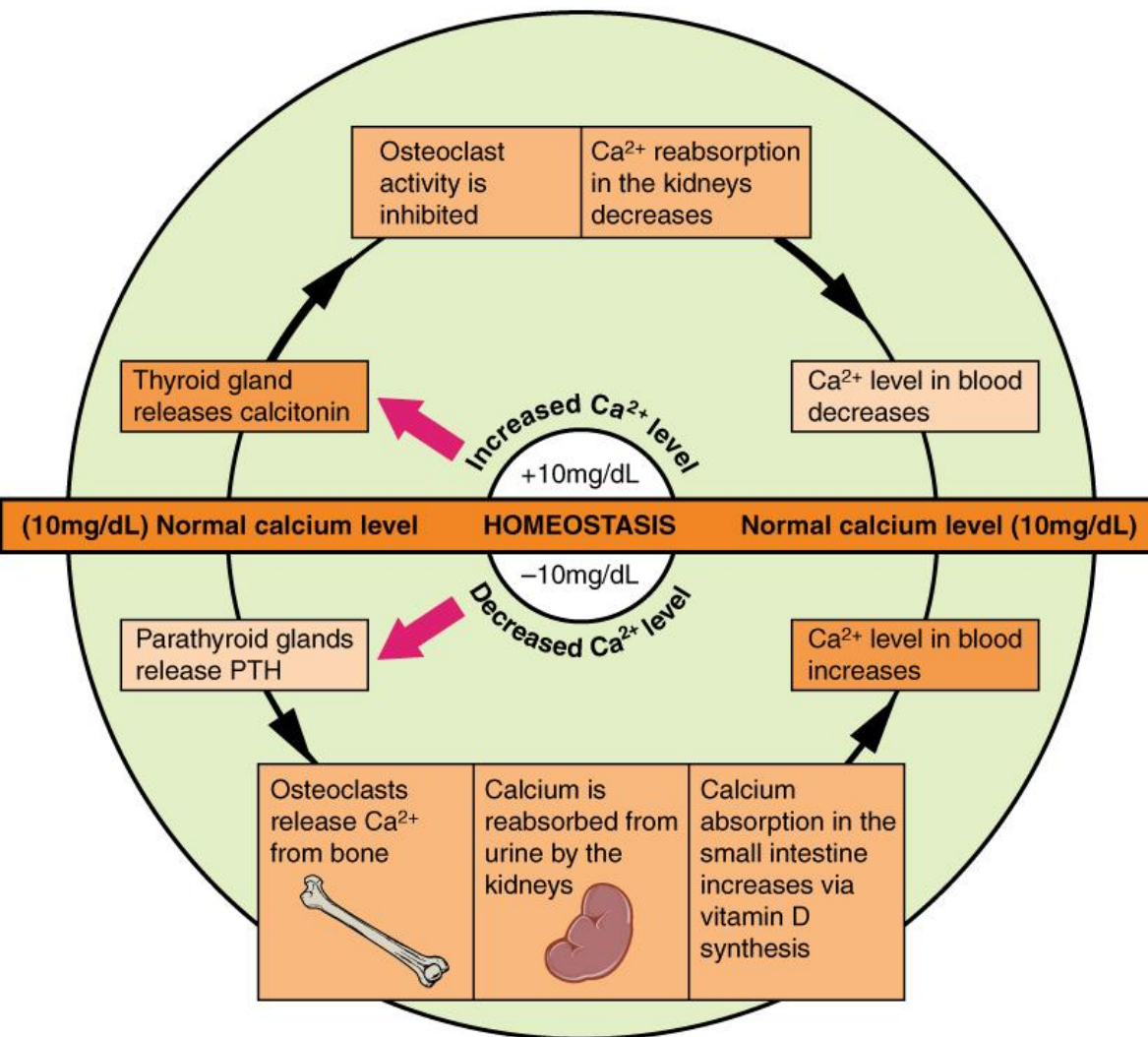
PARATHYROID HORMONE (PTH)

- ✓ Synthesized as **pre-proparathyroid hormone** (115 aa), subsequently cuts by enzymatic with the formation of parathyroid hormone (90 aa) and,
 - **Parathyroid hormone (84 aa)**
 - Sequence 1-34 is highly conserved and appears to be involved in receptor binding
- ✓ Its secretion is stimulated by the **reduction of calcium concentration** in the extracellular fluid and is inhibited **by its increase** and by **1,25-DHCC**
- ✓ Circulating hormone is cleared by the liver and the kidney



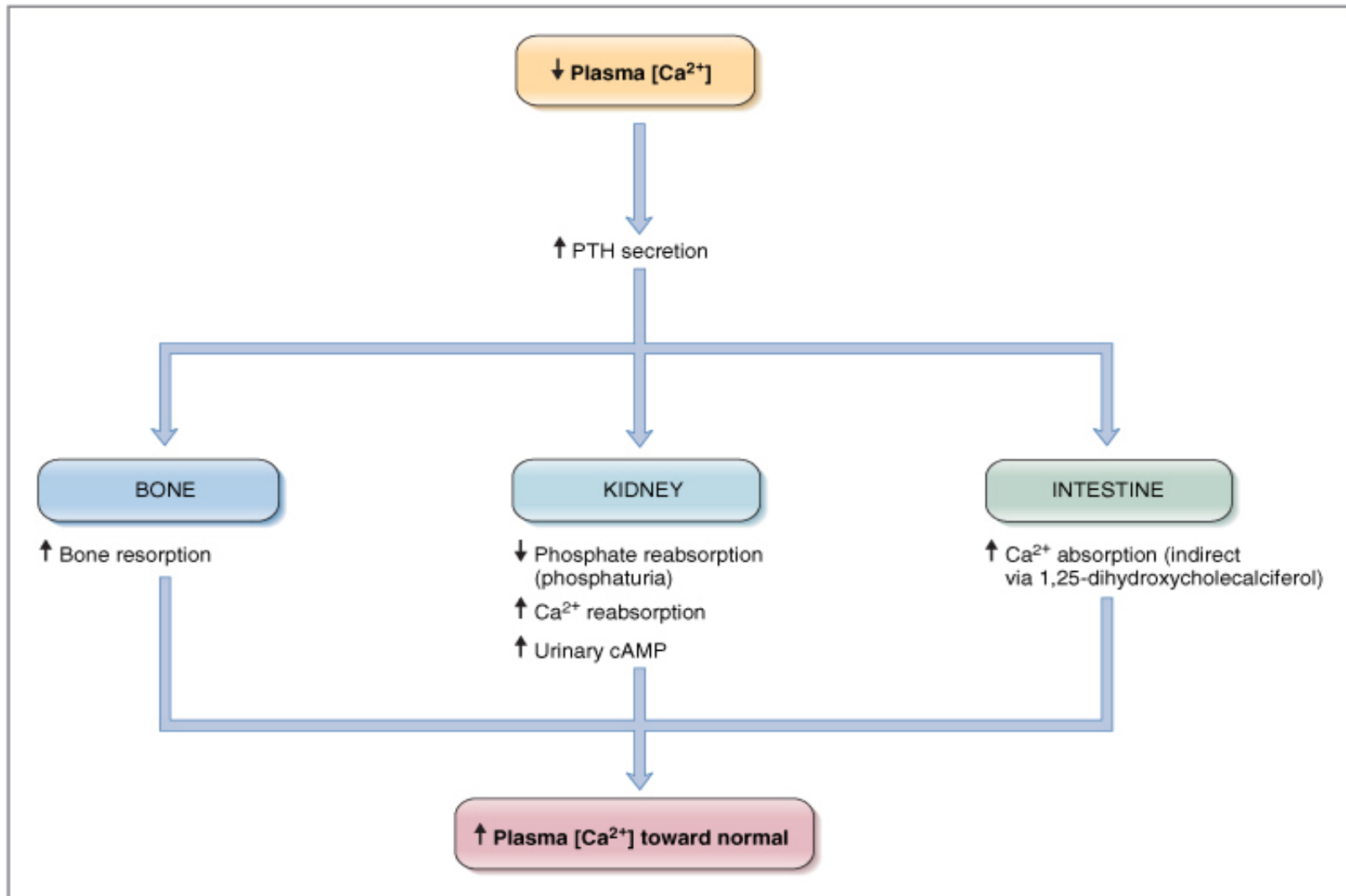
**Parathyroid
glands**

Calcium homeostasis....



PTH EFFECTS

- ✓ **PTH** after interact with the type I receptor (PTH1R);
carried out different activities to increases plasma Ca^{++} at
different tissues:-
- ❖ At the level of **the bone**
 - it determines stimulation of osteoclasts with
resorption of the bone matrix with release of Ca^{2+}
- ❖ At the **Kidney** perform two things
 - it increases the tubular reabsorption of Ca^{++} in distal
convoluted tubule (DCT).
 - It increases the activity of the 1-hydroxylase resulting
in increased synthesis of **1,25-DHCC (vit.D)** in the
kidney



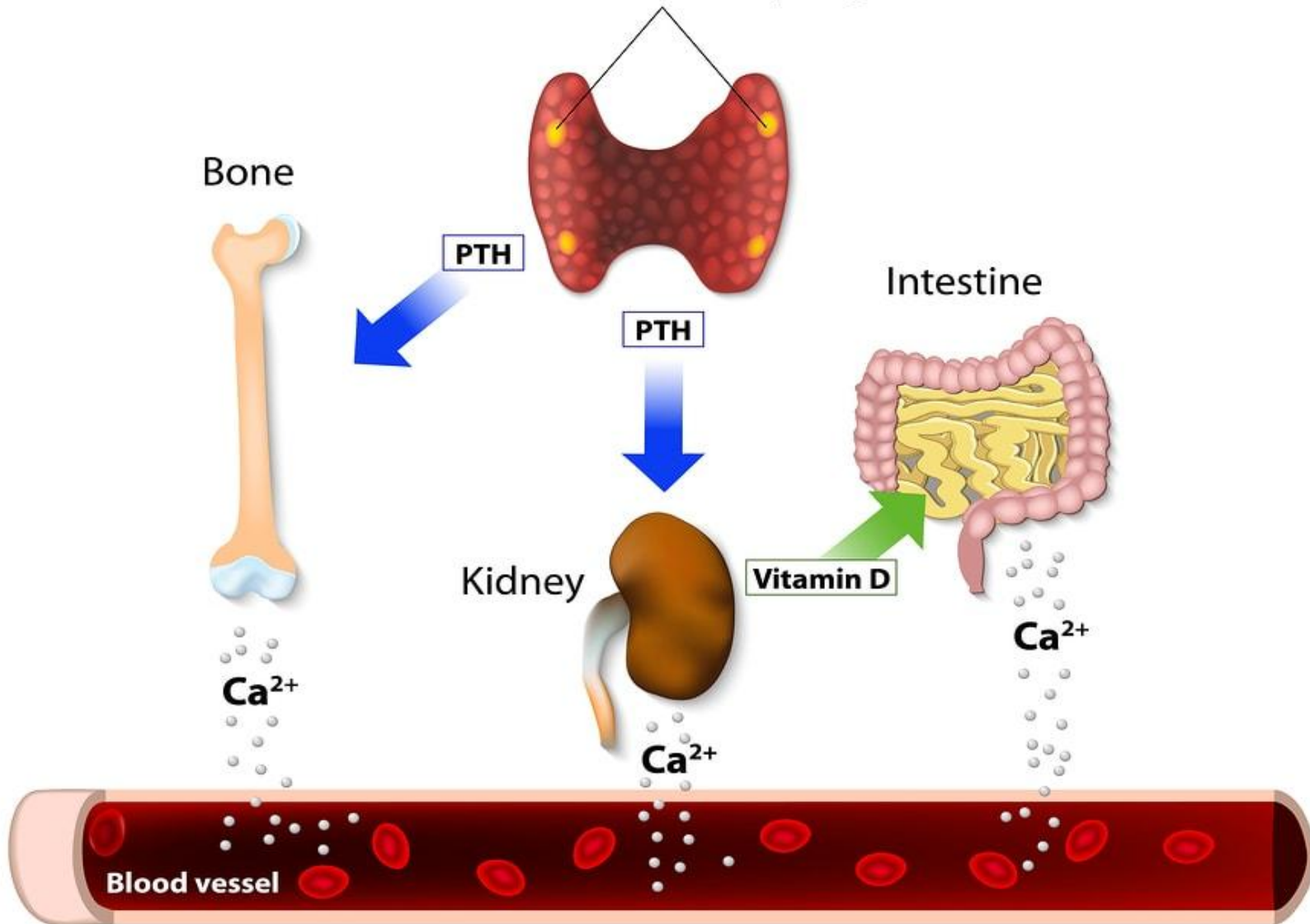
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Figure 9-37 Regulation of PTH secretion and PTH actions on bone, kidney, and intestine. cAMP, Cyclic [adenosine](#)^{Px} monophosphate; PTH, parathyroid hormone.

PARATHYROID GLANDS

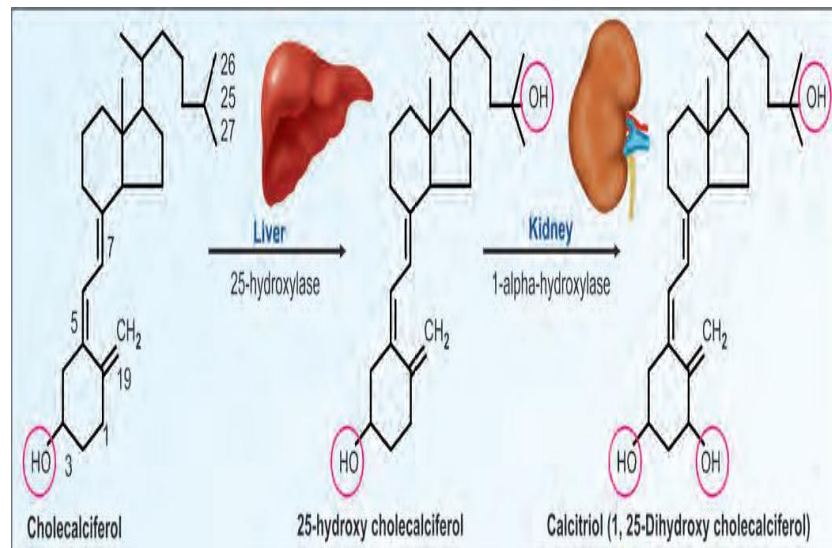
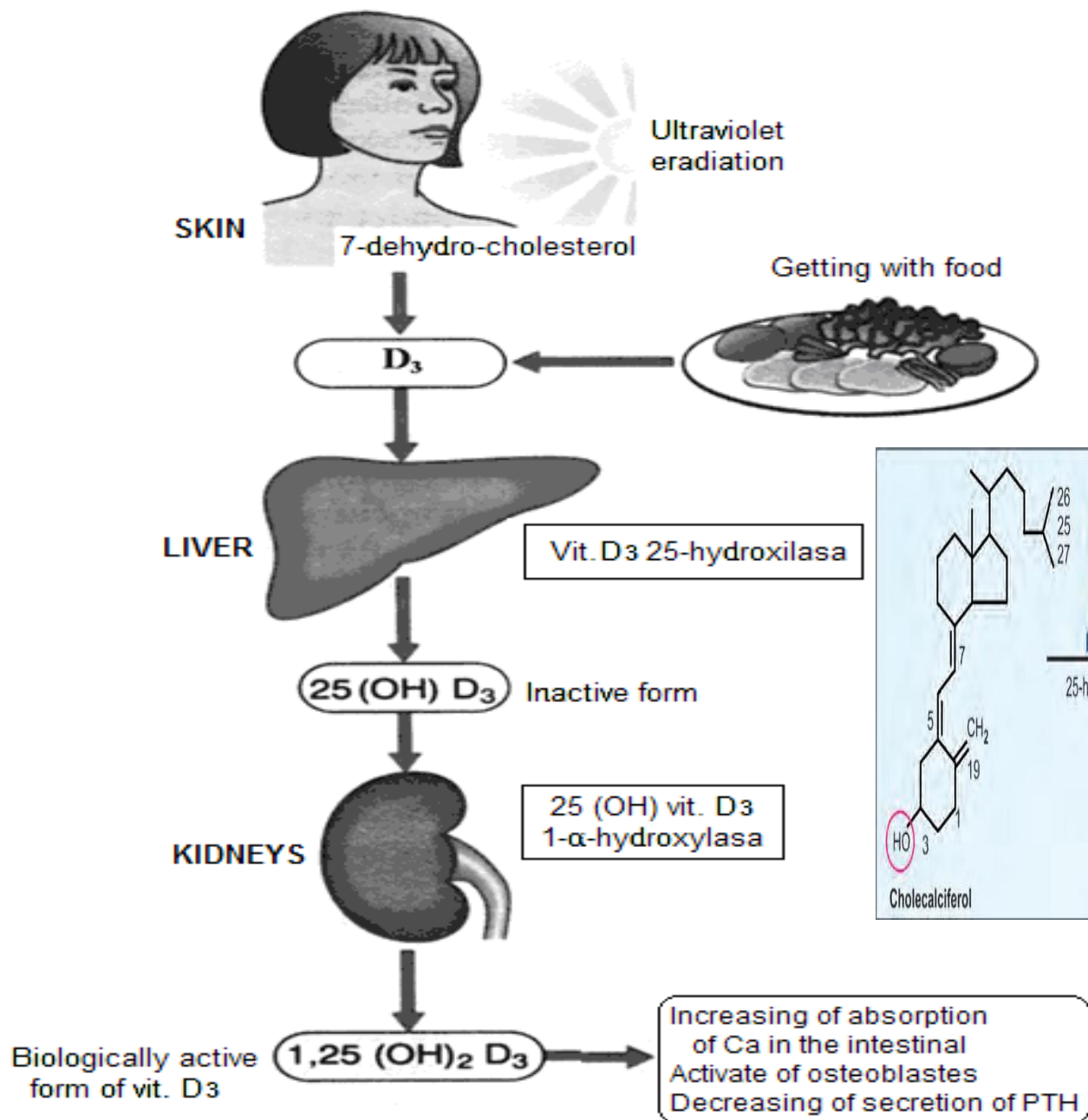
(located on the back of the thyroid gland)





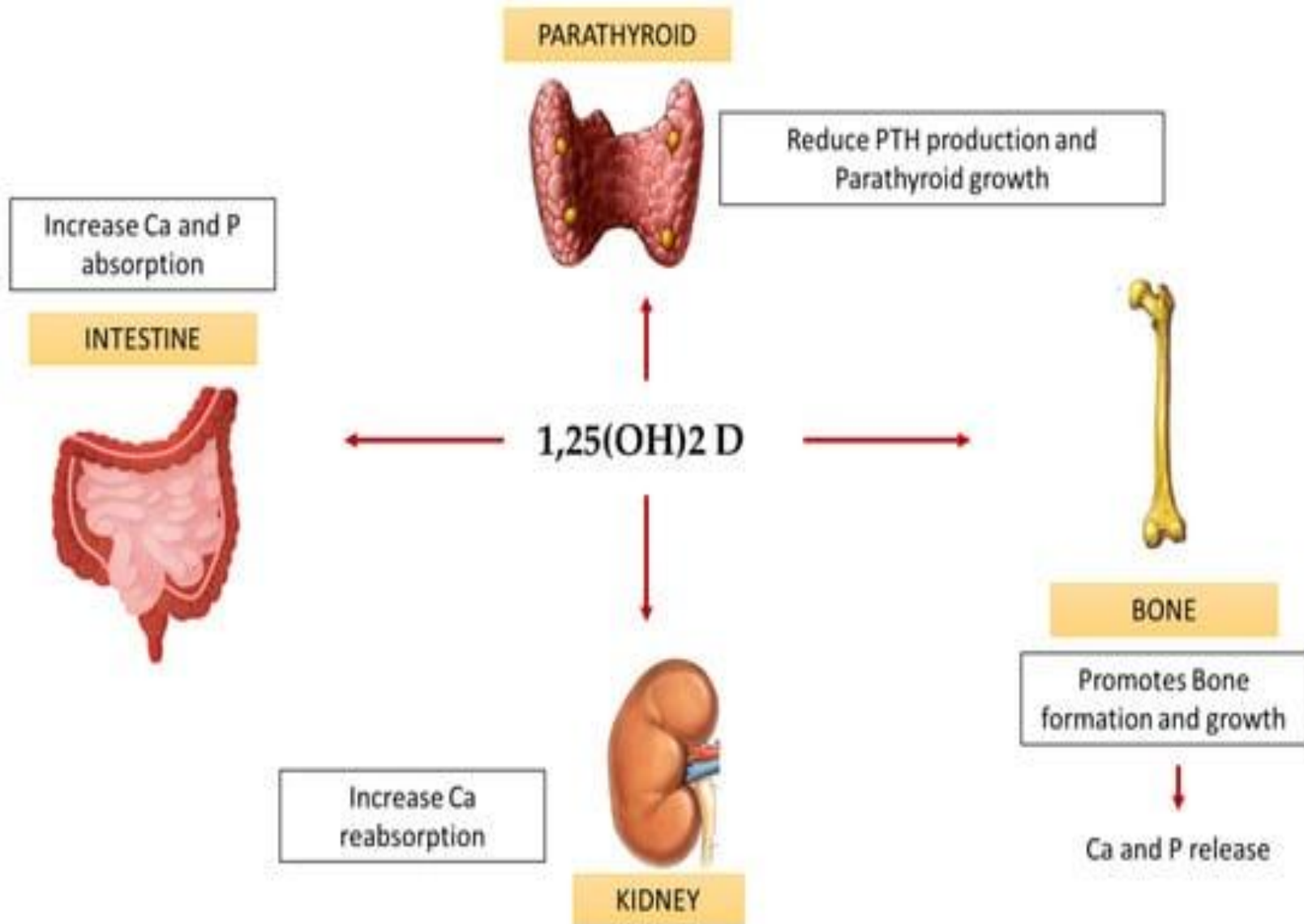
1,25-Dihydroxycholecalciferol/Vitamin D

- ✓ Vitamin D requires activation by **double hydroxylation**
 - The first, **position 25**, is implemented by the liver and is not subject to any control,
 - the second, **position 1**, is activated by the kidney and is controlled by **PTH**
- ✓ The activity of the 1-hydroxylase is increased by PTH and by the reduction of **Ca²⁺** concentration in ECF while it is inhibited by the product of the reaction **(1,25-DHCC)**
- ✓ It is **inactivated** by hydroxylation in **position 24**



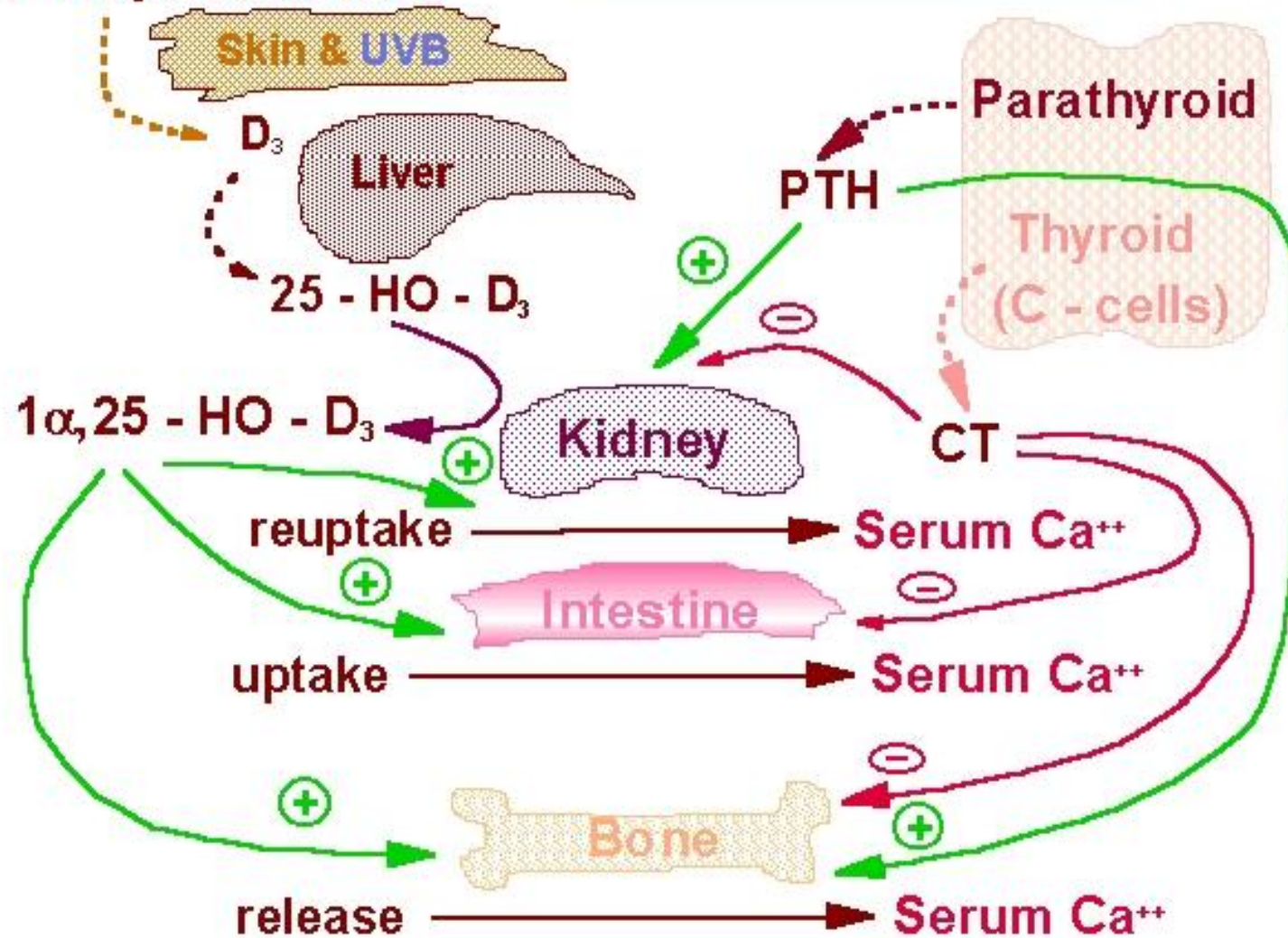
EFFECTS OF VITAMIN D

- ✓ It acts by binding its specific receptor belonging to the nuclear receptor superfamily VDR activation is followed by nuclear translocation and induction of the expression of target genes
- ❖ **Target organs:**
 1. **Intestinal** : increased expression of membrane **Ca²⁺ channels**
→ promotes the absorption of calcium from the intestine
 2. **Bone**: **Mineralization** of the bone is increased by increasing the activity of osteoblasts → increases bone mineral density
 - ✓ Active vitamin D has action on all three types of bone cells
→ osteoblasts, osteoclasts and osteocytes
 3. **Kidney**: increases the reabsorption of **calcium by renal tubules**, therefore it is conserved.



Calcium Homeostasis

7 - dehydrocholesterol



Deficiency of Vitamin D

- ❖ Vitamin D deficiency can lead to various health issues, particularly affecting bone health.
- ❖ The optimal concentration of 25-hydroxy D3 is **> 30ng/mL**, where as **10–19ng/mL** is **deficient**.
- ❖ Rickets
 - the bones of children are undermineralized as a result of poor absorption of calcium → soft bones and deformities
- ❖ Osteomalacia (softens bones)
 - in adults results from demineralization of bone, more common in women & often happens during pregnancy.

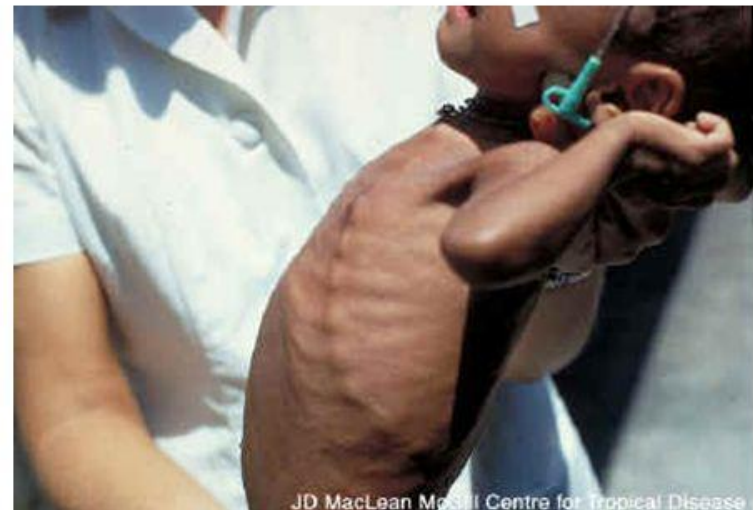
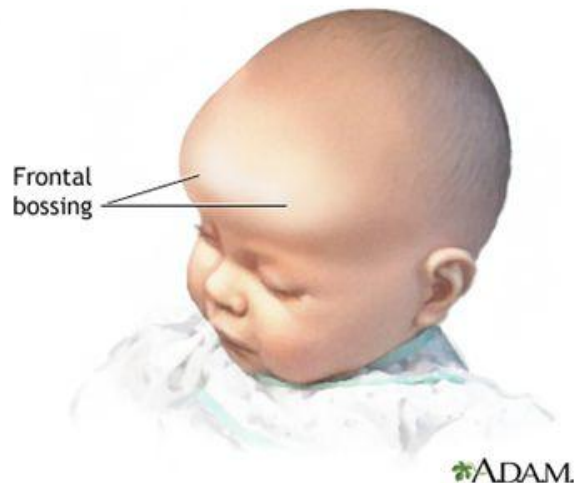
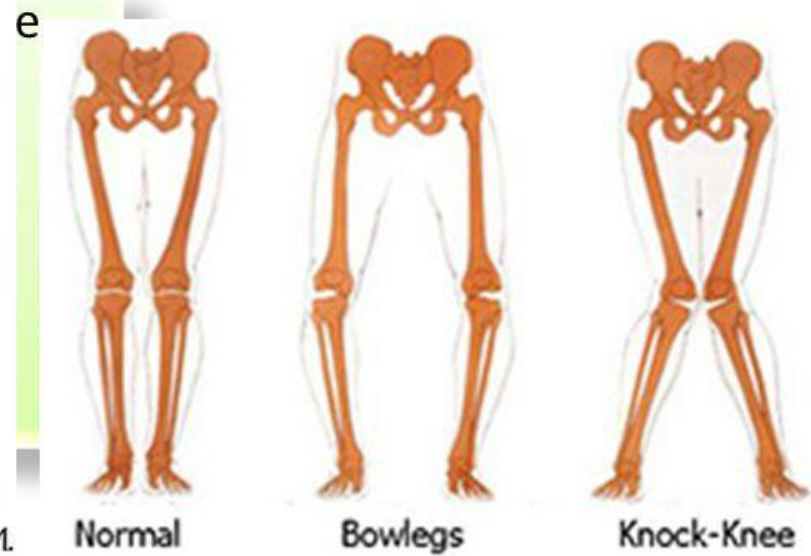
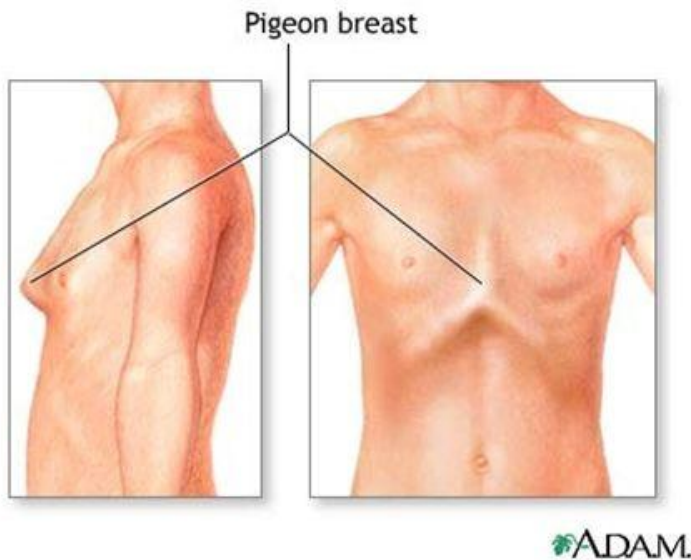
Cause of Vitamin D deficiency

1. Poor sunlight exposure , e.g. ➔ inhabitants of northern latitudes, in people who are bedridden for long periods.
2. Nutritional deficiency of calcium may also produce similar clinical picture.
3. Malabsorption of vitamin (obstructive jaundice and steatorrhea) ➔ High phytate content in diet.
4. Abnormality of vitamin D activation ➔ Liver and renal diseases

Clinical Features of Rickets

- ❖ Rickets, a bone-softening disease in children, presents with a range of clinical features, including skeletal deformities like
 - **bowed legs,**
 - **thickened wrists and ankles**
 - **rachitic rosary (rib swelling).**
- ❖ Other symptoms include bone pain, muscle weakness, poor growth, and dental problems.
- ❖ In severe cases, **seizures** and **fractures** may occur

Clinical manifestations of of Vit D def.



Osteomalacia

- ❖ The term is derived from Greek “**osteon**” = bone; and “**malakia**” = softness.
 - The bones are softened due to insufficient mineralization → **increased osteoporosis**.
 - Patients are more prone to get fractures.
- ❖ The abnormalities in biochemical parameters are a slightly lower serum calcium, and a low serum phosphate.



Feature	Osteoporosis	Osteomalacia
Core Problem	Reduced bone density (porous and weak)	Defective bone mineralization (soft and weak)
Bone Matrix Ratio	Normal mineral-to-matrix ratio, but less of it	Low mineral-to-matrix ratio
Primary Cause	A combination of low peak bone mass and increased bone loss	Vitamin D deficiency (or calcium/phosphate deficiency)
Key Symptom	Often no symptoms until a fracture occurs	Bone pain, muscle weakness, and deformities like bowed legs
Fracture Type	Fractures from minor events like a fall	Often associated with stress fractures in softened bones

BONE PROTEINS

BONE PROTEINS

- Bone proteins play a crucial role in maintaining the structure, strength, and health of our bones.
- They make up about **50% of the bone volume** and essential framework for the **mineral deposits**.
- There are two main categories of bone proteins:
 - i. Collagen:** make up about 90% of the organic matrix.
 - ✓ D/t types, **type I collagen** primarily found in bone
 - ii. Non-collagenous proteins:** These proteins make up the remaining 10% of the organic matrix and
 - ✓ **bone dev't, mineralization, and maintenance.**

Collagen

- **Most abundant** protein in human body (25% of protein mass)
 - ✓ It provides structural support to the extracellular space of connective tissues.
 - ✓ Due to its **rigidity and resistance to stretching**, it is the perfect matrix for skin, tendons, bones, and ligaments.
 - ✓ Typical Collagen is long, rigid structure in w/c **three polypeptides** (“ α -chains”) are wound around one another in a rope-like **triple helix**

Collagen.....

- ❖ Has structural functions in d/t organs w/c dictates its type & organization:-
 - **In ECM or vitreous humor of eye:**
 - ✓ dispersed as a gel that gives structural support
 - **In cornea of eye:**
 - ✓ stacked so as to transmit light with a minimum of scattering
 - **In tendons:**
 - ✓ bundled in tight, parallel fibers that provide great strength
 - **In bone:**
 - ✓ occurs as fibers arranged at angle to each other so as to resist mechanical shear from any direction

Types of Collagen

- ✓ There are more than 25 collagen types
- ✓ organized into three groups, based on their location & functions

Group-I: Fibril-forming collagens

- Have rope-like triple helix structure
- **Collagen type I, II, & III**
 - ✓ **Type I collagen fibers:** Found in supporting elements of high tensile strength (Skin, bone, tendon, blood vessels & cornea)
 - ✓ **Type II collagen fibers:** restricted to cartilaginous structure (Cartilage, intervertebral disk, vitreous body)
 - ✓ **Type III collagen fibers:** are prevalent in more distensible tissues (Blood vessels, fetal skin)

Types of Collagen.....

Group-II: Network-forming collagens:

- Form 3-D mesh, rather than distinct fibrils
- **Collagen type IV & VII**
 - ✓ **Type IV collagen:** assembles into a meshwork that constitutes a major part of glomerular basement membranes of Kidney
 - ✓ **Type VII collagen:** assembles into a meshwork beneath stratified squamous epithelia

Group III: Fibril-associated collagens:

- Bind to surface of collagen fibrils, linking these fibrils to one another & to other components in ECM
- **Collagen type IX & XII**

TYPE	TISSUE DISTRIBUTION
	Fibril-forming
I	Skin, bone, tendon, blood vessels, cornea
II	Cartilage, intervertebral disk, vitreous body
III	Blood vessels, fetal skin
	Network-forming
IV	Basement membrane
VII	Beneath stratified squamous epithelia
	Fibril-associated
IX	Cartilage
XII	Tendon, ligaments, some other tissues

Structure of Collagen

- Its triple-helical structure that places many of its amino acid side chains on surface of helix
- Collagen is rich in **Proline** & **Glycine** amino acids
 - **Pro** facilitates formation of helical conformation of each α -chain b/c its ring structure causes “kinks” in peptide chain
 - **Gly** founds in every third position of polypeptide chain
 - **Gly** fits into restricted spaces where three chains of helix come together
 - Presence of Pro & Gly dictates that helical conformation of α -chain **cannot** be **α -helix**

Structure of Collagen.....

- ❖ Repeating sequence of amino acids in triple helix **Gly-X-Y** Where:

X = Frequently Proline

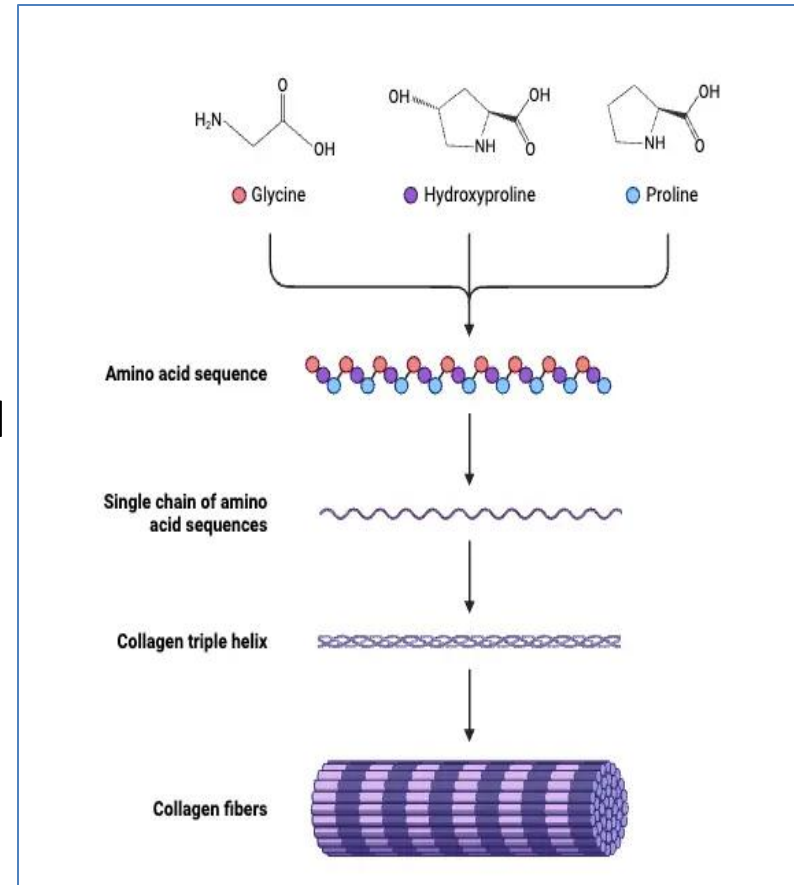
Y = Often Hydroxyproline or sometimes, Hydroxylsine

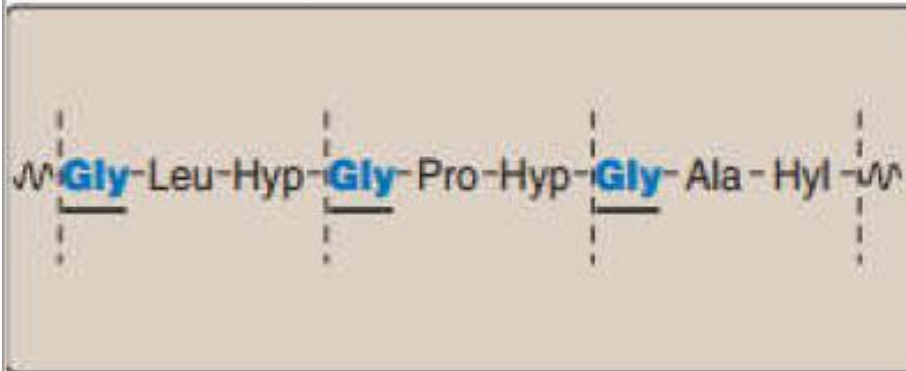
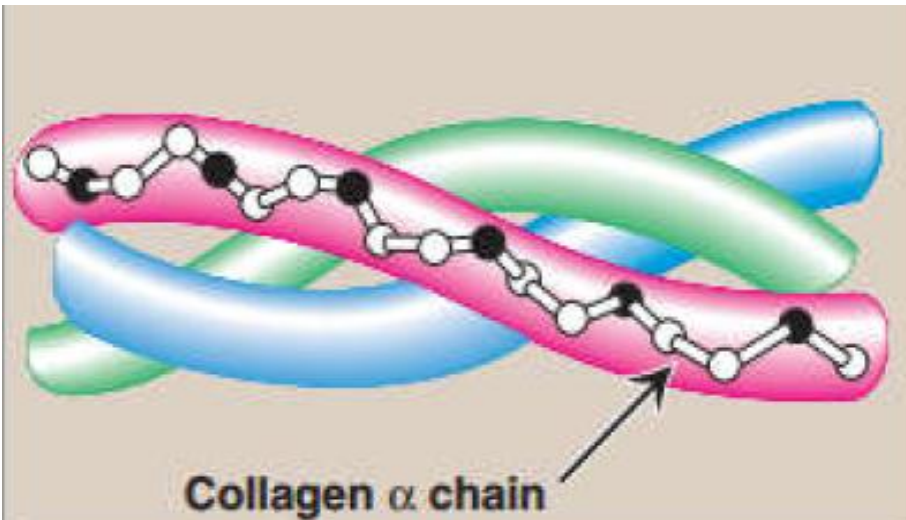
- ❖ **Hydroxyproline (Hyp):**

- ✓ Results from hydroxylation of some **Proline** residue during post translational modification polypeptide chains
- ✓ Important in stabilizing triple-helical structure of collagen b/c it maximizes interchain hydrogen bond formation

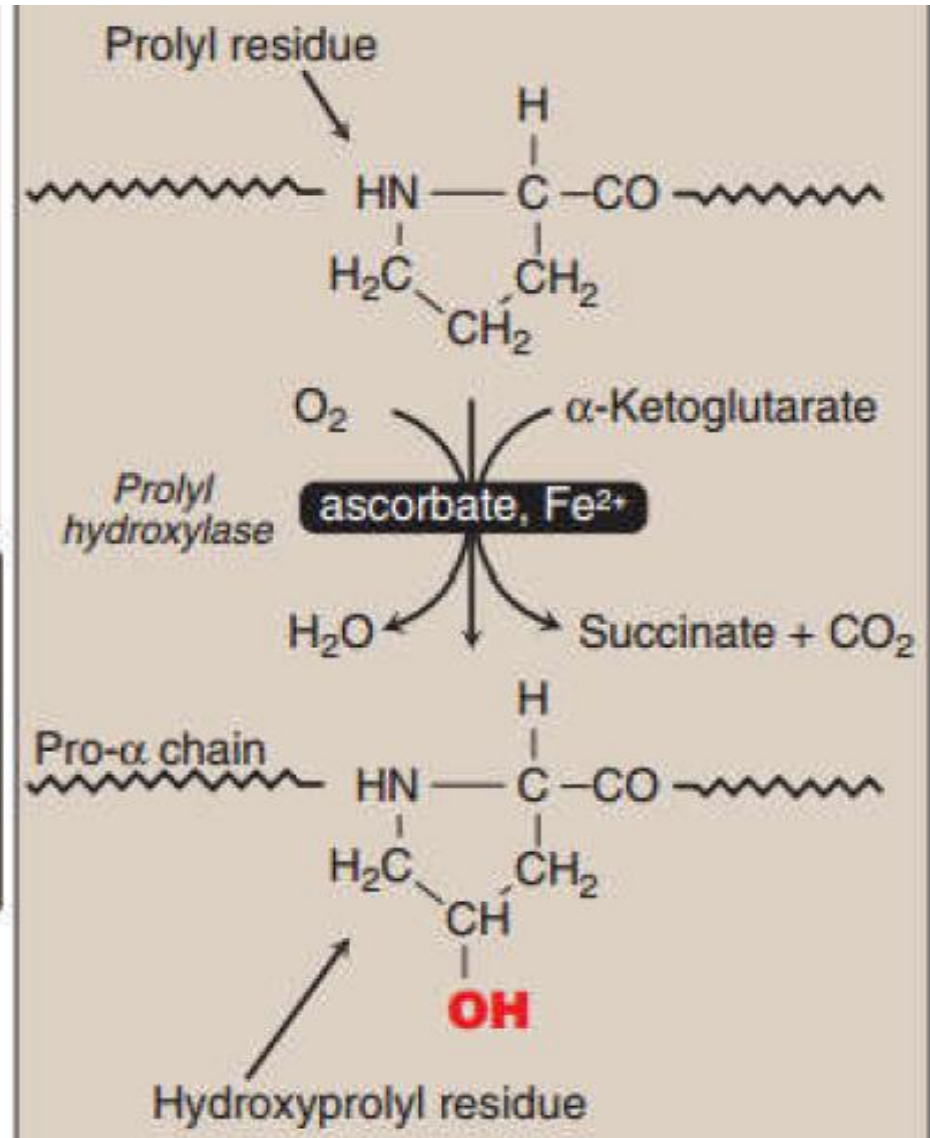
- ❖ **Hydroxylsine (Hyl)**

- ✓ Results from hydroxylation of some **Lysine** residue during post-translational modification polypeptide chains





Structure of Collagen



Hydroxylation reaction of Proline

Biosynthesis of Collagen

- ✓ Collagen is initially synthesized as a larger precursor polypeptide, **procollagen**.
- ✓ Numerous prolyl and lysyl residues of procollagen are **hydroxylated**
- ✓ After enzymatic modification, mature collagen monomers **aggregate & become cross-linked** to form collagen fibers
- ✓ Collagen synthesis starts with **activation genes** for α -peptide chains in DNA
- ✓ There are multiple genes to synthesize d/t types of α -peptide chains

Biosynthesis of Collagen.....

➤ Steps of Collagen Biosynthesis

- Collagen is one of many proteins that normally function outside of cells
- Collagen biosynthesis involves **two steps**:
 1. Intracellular steps
 2. Extracellular steps

Biosynthesis of Collagen.....



1. Intracellular steps

step1:- Formation of pro- α chains

- ✓ Newly synthesized polypeptide precursors of **α - chains (prepro- α chains)**
 - Contain special amino acid sequence
 - Rapidly cleaved in RER to yield a precursor of collagen called **pro- α chain**

step2: Hydroxylation:

- ✓ Within lumen of RER Pro- α chains are processed by number of enzymatic steps **Pro** & **Lys** residues found in **Y-position** of **–Gly–X–Y–** sequence undergo hydroxylation by Vitamin C dependent enzymes (Prolyl hydroxylase & Lysyl hydroxylase)

Biosynthesis of Collagen.....

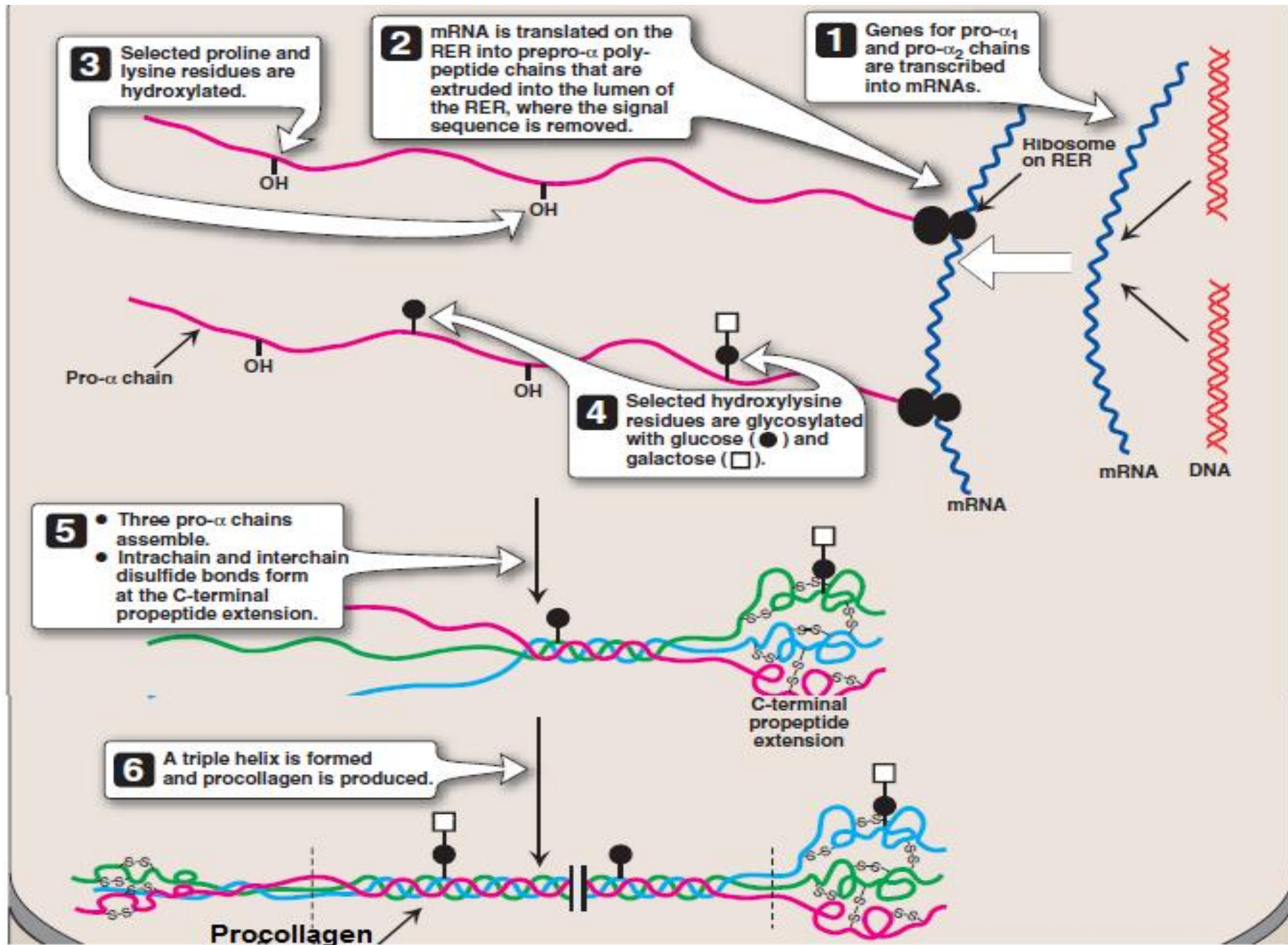


Step3:- Glycosylation:

- ✓ Some Hydroxylysine residues are modified by enzymatic glycosylation with **glucose** or **glucosyl-galactose**

Step4:-Assembly & secretion

- ✓ After hydroxylation & glycosylation:
 - ❖ **Interchain disulfide bonds** formed b/n three pro- α chains at C-terminal & bring three pro- α chains into alignment favorable for **helix formation**
 - ❖ **Procollagen** (Precursor of collagen that has central region of triple helix flanked by non-helical N- & C-terminal extensions) is formed
 - ❖ The procollagen molecules move through the **Golgi apparatus**, where they are packaged in secretory vesicles.
 - ❖ The vesicles fuse with the cell membrane, causing the release of procollagen molecules into the **extracellular space**



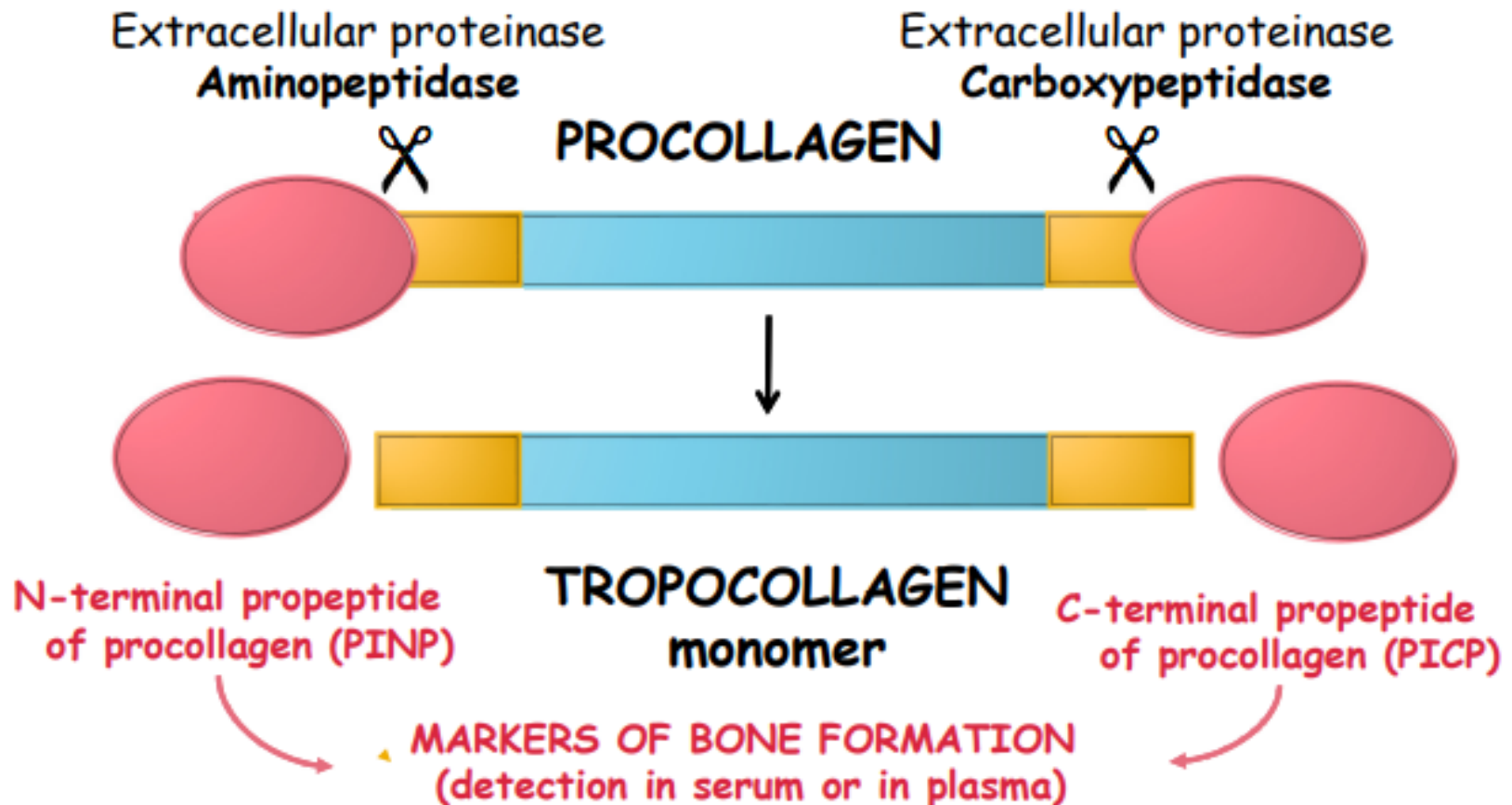
Biosynthesis of Collagen.....



2. Extracellular Steps

- I. Extracellular cleavage of procollagen:** N- & C-procollagen peptidases remove terminal **propeptides**, releasing triple-helical molecule ➔ **Tropocollagen**
- II. Formation of collagen fibrils:** by each tropocollagen molecules overlapping its neighbor
- III. Cross-link formation:** to form ➔ Mature collagen fibers

Cleaving of the propeptides



Assembling of collagen fibrils

Polymerisation

Tropocollagen



Collagen fibrils



Collagen fibers

Scurvy



- In **ascorbic acid (Vit. C)** deficiency hydroxylating enzymes, prolyl hydroxylase & lysyl hydroxylase, are unable to function
- Lack of prolyl & lysyl hydroxylation
 - Inter chain H-bond formation is impaired ➔ Formation of unstable triple helix
 - Collagen fibrils cannot be cross-linked, greatly decreasing tensile strength of assembled fiber
- Patients often show bruises on limbs as result of subcutaneous extravasation of blood due to capillary fragility

»Scurvy:

➤ Deficiency of ascorbic acid

— Results:

- Disease characterized by sore & spongy gums, loose teeth, fragile blood vessels, swollen joints, & anemia



Osteogenesis Imperfecta

- ✓ Mutations in gene for Gly residues in **pro- α 1 or pro- α 2** chains of type I collagen cause replacement of Gly residues (–Gly–X–Y–) by amino acids with bulky side chains
- ✓ Resultant structurally abnormal pro- α chains prevent formation of required triple-helical conformation
- ✓ Final develop **Osteogenesis Imperfecta** (Brittle Bone syndrome)
- ✓ This disorders distinguished by bones that easily bend & fracture

Osteogenesis Imperfecta.....

❖ Type-I: Osteogenesis imperfecta tarda

- ✓ Due to moderate mutation leading to decreased production of $\alpha 1$ & $\alpha 2$ chains of type-I collagen
- ✓ Moderate form Osteogenesis imperfecta in w/c baby will be born (defect is compatible with life)
- ✓ Secondary to minor trauma, infant gets multiple bone fractures
- ✓ Patients may have **blue sclera** due to tissue thinning that allows underlying pigment to show through

❖ Type-II: Osteogenesis imperfecta Congenita

- ✓ Severe form of osteogenesis imperfecta in w/c product of conception suffer from multiple mutation, Intra-uterine death

Ehlers-Danlos syndrome (EDS)

- ✓ Heterogeneous group of disease in w/c extracellular processing of collagen is disturbed
- ✓ Causes:
 - Inherited deficiency of extracellular collagen-processing enzymes
- ✓ Defects in collagen type I fibrils processing → Fragile, **stretchy skin** & loose joints



Muscle Metabolism



- Metabolism in the cells of skeletal muscle, **myocytes**, is specialized to generate ATP as the immediate source of energy for contraction.
- Moreover, **skeletal muscle is adapted** to do its mechanical work in an intermittent fashion, on demand.
- Sometimes skeletal muscles must work at their maximum capacity for a **short time, as in a 100m sprint**; at other times **more prolonged work is required**, as in running a marathon or extended physical labor
- There are **two general classes of muscle tissue**, which differ in physiological role and fuel utilization

Classes of muscle tissue



- I. **Slow-twitch muscle**, also called **red muscle**, provides relatively low tension but is highly resistant to fatigue
 - ✓ It produces ATP by the relatively slow but steady process of oxidative phosphorylation.
 - ✓ **rich in mitochondria** and is served by very dense networks of BVs, which bring the oxygen essential to ATP production.
 - ✓ **Cytochromes in mitochondria** & Hb in blood → **red color**
- II. **Fast-twitch muscle**, or **white muscle**, has fewer mitochondria and is less well supplied with BVs, but it can develop greater tension, and do so faster.
 - ✓ quick for fatigue → **when active it uses ATP faster than it can replace it.**

Muscle Metabolism



- ❖ Muscles primarily rely on three major pathways to generate the energy molecule:

1. Phosphagen System:

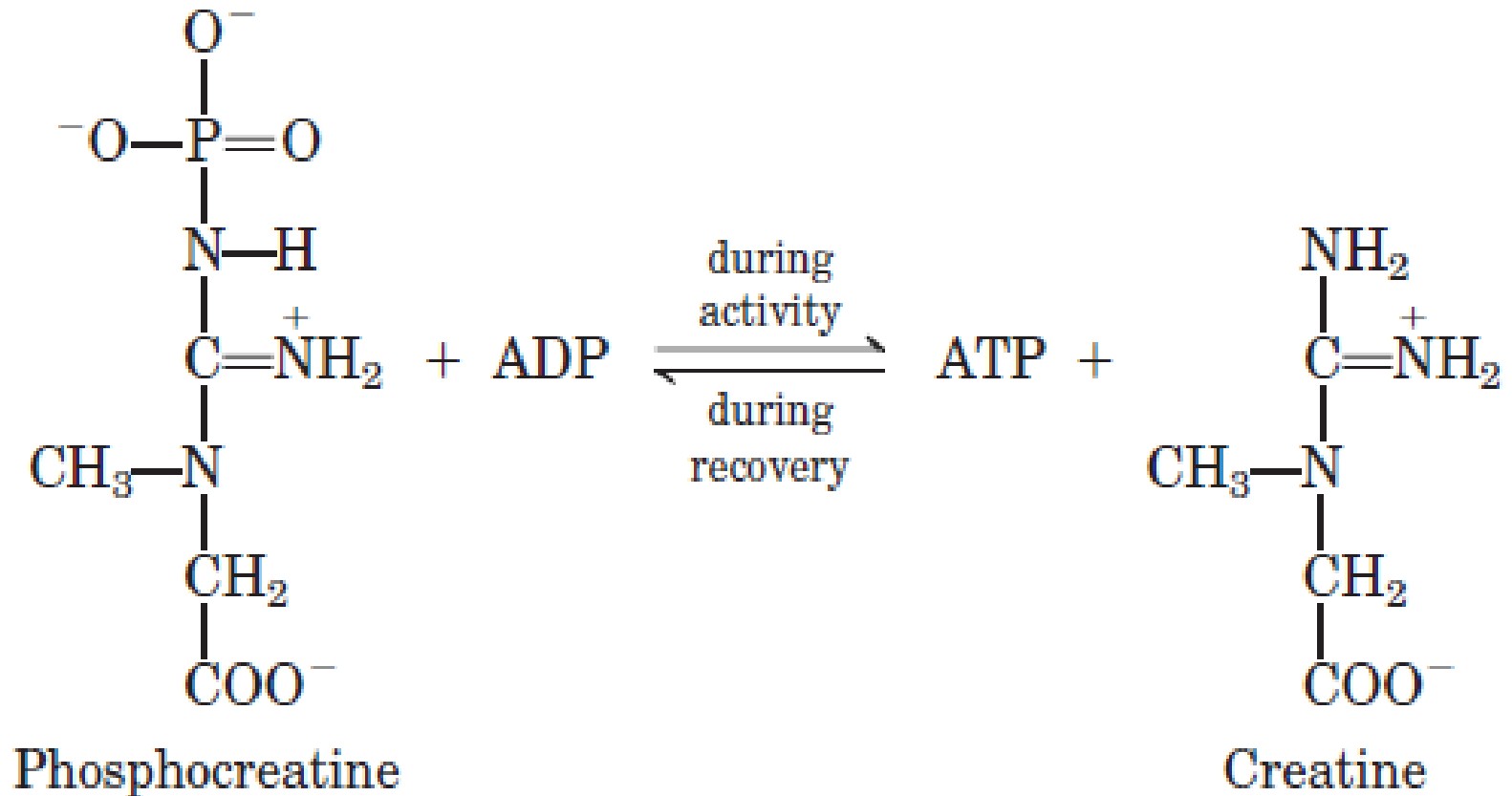
- This system uses **creatine phosphate (CP)**, a readily available high-energy molecule stored in muscles.
- It provides immediate but limited ATP, making it ideal for short bursts of intense exercise (e.g., sprinting).
- When CP is depleted, the body shifts to other pathways.

Phosphocreatine

- known as **Creatine phosphate**, an important energy stored in the skeletal muscle.
- **Creatine is synthesized in the liver** and transported to the muscle cells & phosphorylated → **Creatine phosphate**.
- This reaction occurs in the **sarcoplasm**.
- It is subject to depletion during extended periods of contraction (intense effort).
- **Rephosphorylation** of Creatine occurs at the mit. membrane.



Formation of phosphocreatine



Muscle Metabolism.....



2. Glycolytic System:

- This system breaks down **glycogen**, stored carbohydrates within muscles, into **glucose** for energy.
- It offers more sustained energy than the phosphagen system but can lead to lactate production, contributing to muscle fatigue.
- High-intensity exercise primarily relies on this pathway.

Muscle Metabolism.....



3. Oxidative System:

- This system utilizes **oxygen** to break down **fatty acids** and **glucose** (derived from blood sugar or glycogen) for ATP production.
- It provides the most energy but takes longer to activate, making it dominant during low-intensity, prolonged exercise (e.g., endurance activities).

Fuel utilization muscle tissue



- **In resting muscle**, the primary fuels are
 - fatty acids from adipose tissue & ketone bodies from liver.
 - ✓ These are oxidized and degraded to yield acetyl-CoA, which enters the **citric acid cycle** for oxidation to CO₂.
 - ✓ The ensuing transfer of electrons to O₂ provides the energy for ATP synthesis by oxidative phosphorylation.

- **Moderately active muscle** uses
 - blood glucose in addition to fatty acids & ketone bodies.
 - ✓ The glucose is phosphorylated, then degraded by glycolysis to pyruvate → **acetyl-CoA** and oxidized via:
 - citric acid cycle and oxidative phosphorylation

Fuel utilizationcont'd



- **In maximally active fast-twitch muscles(vigorous exercises)**, the demand for ATP is so great that;
 - ✓ the blood flow cannot provide O_2 , so ATP cannot synthesis by **oxidative phosphorylation**.
 - ✓ Under these conditions, **stored muscle glycogen** is broken down to **lactate** by anaerobic respiration.

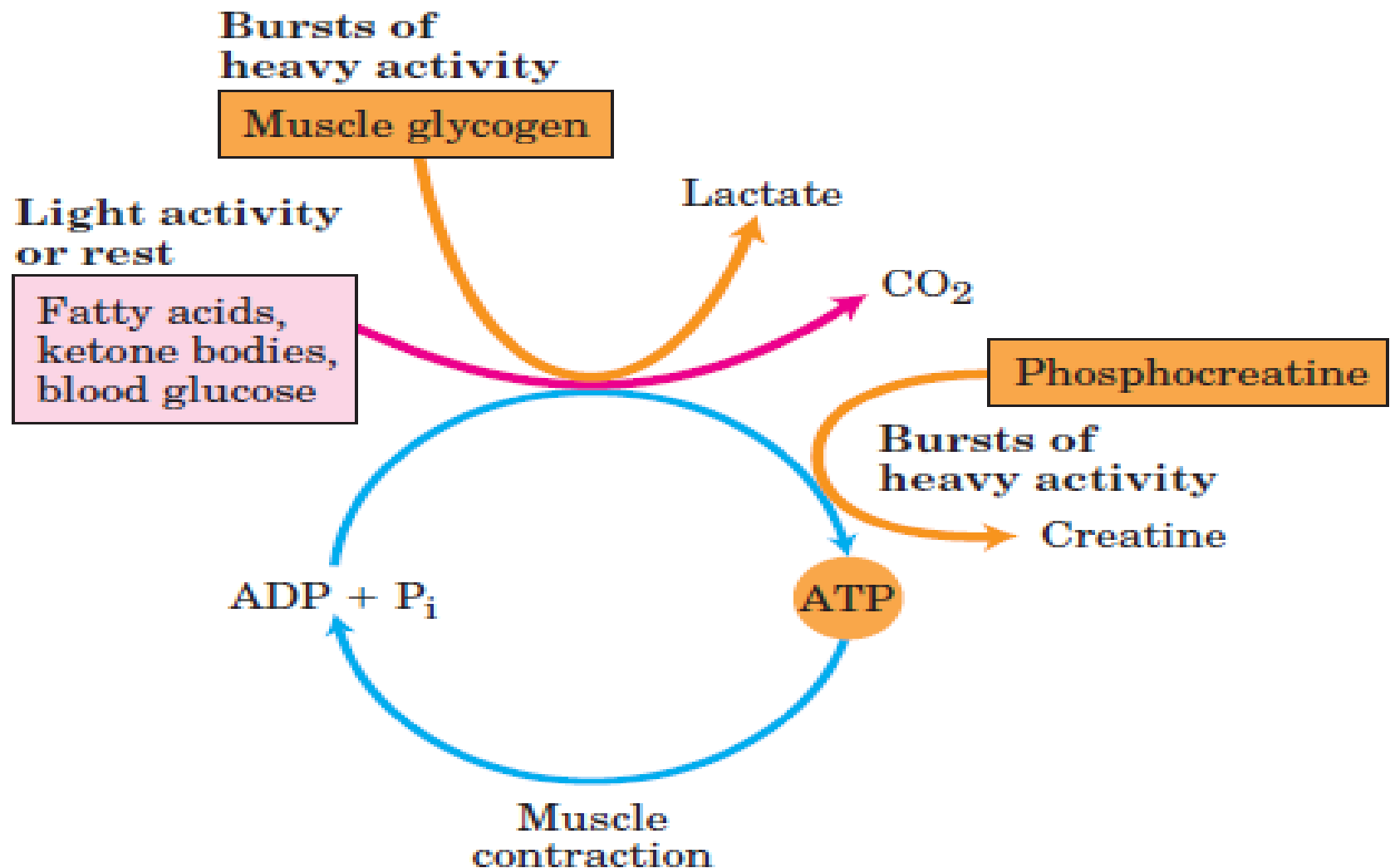


FIGURE 23-17 Energy sources for muscle contraction. Different fuels are used for ATP synthesis during bursts of heavy activity and during light activity or rest. Phosphocreatine can rapidly supply ATP.

Thank You!!!